

SkillSync - AI Career Growth & Upskilling Assistant

End-to-End User Flow, Architecture, and Implementation Plan

1. Project Overview

SkillSync is an AI-powered career growth and upskilling platform that provides personalized roadmaps, project ideas, and resume generation based on user skills and goals. This document explains the complete user flow, backend data handling, AI integrations, and development roadmap for building the system using the MERN stack (MongoDB, Express.js, React.js, Node.js).

2. Visual User Flow Diagram

```
[Landing Page]
  ?
[Login / Register] -> [JWT Auth]
  ?
[Onboarding Wizard]
  ?? Choose Career Goal
  ?? Upload Resume or Enter Skills
  ?
[AI Resume Parser -> Extract Skills]
  ?
[AI Roadmap Generator -> Personalized Roadmap]
  ?
[AI Project Suggestions]
  ?
[Dashboard]
  ?? Skills & Roadmap Progress
  ?? Resume Builder
  ?? Job Market Insights
  ?? AI Career Chat Assistant
  ?
[Continuous Learning & Tracking]
```

3. Feature-by-Feature Implementation Roadmap

Phase 1 - Project Setup:

- ? Create /client (Vite React) and /server (Express.js)
- ? Install dependencies and connect MongoDB Atlas
- ? Test backend route '/api/test'

Phase 2 - Authentication:

- ? JWT-based login/register with bcrypt
- ? Redirect user to onboarding after login

Phase 3 - Onboarding:

- ? Choose goal, upload resume or add skills manually
- ? AI extracts and confirms user skills

Phase 4 - AI Roadmap Generator:

- ? Generate personalized 3-phase learning plan using OpenAI/Gemini

Phase 5 - Project Recommender:

- ? Suggest AI-based projects for practice

Phase 6 - Dashboard:

- ? Unified dashboard showing skills, progress, and insights

Phase 7 - AI Chat Assistant:

- ? Chat interface powered by LLM for career queries

Phase 8 - Resume Builder:

- ? Generate and download tailored resumes & cover letters

Phase 9 - Admin Panel:

- ? Manage users, track analytics, and job trends

Phase 10 - Deployment:

- ? Frontend -> Vercel | Backend -> Render | Database -> MongoDB Atlas

4. System Architecture Overview

[Frontend: React + Vite]

? (Axios)

[Backend: Express.js + Node.js]

?

[Database: MongoDB Atlas]

?

[AI Layer: OpenAI / Gemini API]

?

[External APIs: LinkedIn, Job Market APIs]

5. Frontend <-> Backend Data Flow

1. User registers/logs in -> /api/auth/login
2. Upload resume -> /api/resume/upload (AI skill extraction)
3. Generate roadmap -> /api/ai/roadmap
4. Get projects -> /api/ai/projects
5. AI Chat -> /api/ai/chat
6. Resume Builder -> /api/ai/resume
7. Job Trends -> /api/jobs/trends
8. Update progress -> /api/user/roadmap/update

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